

Fusion-Based Uncertainty Quantification for Safety-Gated Retrieval-Augmented Route Assistance

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Abstract—As wrong outputs might influence subsequent decision-making, retrieval-augmented generation (RAG) systems are being investigated more and more for route-assistance and road-notice question answering. While retrieval enhances large language models’ (LLMs’) grounding, generating hallucination and retrieval mismatch both lead to doubt. We introduce a fusion-based uncertainty quantification (UQ) approach that creates a single trust score by combining token-level white-box creation entropy with normalized retrieval similarity. A three-tiered decision policy (green, amber, and red) for operational gating is supported by the fused score once it has been mapped using a scaled sigmoid transformation. We assess the method using risk-coverage, discrimination, and calibration measures on a 50-query labeled route-assistance dataset. Whereas a Polygraph-only uncertainty signal shows high calibration (ECE = 0.0404) but generates compressed trust distributions that are inappropriate for gating, the fused technique obtains AUPRC = 0.6690 and permits non-zero operational coverage at predetermined decision thresholds. The results suggest that merging retrieval and generation uncertainty produces a more operationally usable confidence measure than single-source estimators on the evaluated dataset, even though the fused signal’s calibration is still modest (ECE = 0.2077). These results establish fusion-based UQ as a workable prototype method for RAG systems that are sensitive to uncertainty, with explicit guidelines for validation and calibration improvement on a broader scale. All results are reported on a 50-query prototype dataset and should be interpreted as exploratory.

Index Terms—Uncertainty Quantification, Retrieval-Augmented Generation, Trust Calibration, Route Assistance, Risk-Coverage Analysis

I. INTRODUCTION

A. Motivation and Research Goals

Responses from Large Language Models (LLMs) used in route-assistance applications need to be dependable and considerate of safety. Retrieval-Augmented Generation (RAG) enhances factual grounding, but it also adds sources of uncertainty from generated variability and retrieval failures. Erroneous outputs might result in dangerous routing choices in navigation scenarios. As a result, uncertainty estimation needs to control operational behavior in addition to ranking replies.

This work addresses the following research question:

How can multi-source uncertainty in RAG systems be fused into a normalized trust score that enables safety-aware decision gating in route-assistance applications?

B. Project Overview

We implement a modular RAG pipeline that extracts:

- Retrieval confidence from similarity scores,
- White-box token-level entropy from the generator,
- Optional LM-Polygraph signals.

These signals are normalized and fused via a weighted linear combination followed by sigmoid scaling:

$$\tau = \sigma(6(\alpha G + \beta R - 0.5)) \quad (1)$$

where $\alpha = 0.4$, $\beta = 0.6$, and $\tau \in [0, 1]$ denotes trust.

The trust score drives a three-tier safety gate:

GREEN: $\tau \geq 0.8$

AMBER: $0.6 \leq \tau < 0.8$

RED: $\tau < 0.6$

C. Key Reference Work

- **Shorinwa et al.** Survey on LLM UQ (arXiv:2412.05563).
- **Sharma** RAG Systems Survey (arXiv:2506.00054).
- **Bouchard & Chauhan** UQLM Ensemble Framework (arXiv:2504.19254).

II. BACKGROUND AND CONCEPTS

A. Uncertainty in RAG

RAG models compute:

$$P(y|x) = \sum_{i=1}^k P(y|x, d_i)P(d_i|x) \quad (2)$$

Uncertainty arises from both $P(d_i|x)$ (retrieval reliability) and $P(y|x, d_i)$ (generation uncertainty).

B. White-Box Token Entropy

For token t_i with distribution p_{ij} :

$$H_i = - \sum_j p_{ij} \log p_{ij} \quad (3)$$

Mean entropy:

$$\bar{H} = \frac{1}{n} \sum_{i=1}^n H_i \quad (4)$$

Entropy is normalized to form generation confidence:

$$G = 1 - \frac{\bar{H} - H_{\min}}{H_{\max} - H_{\min}} \quad (5)$$

C. Retrieval Confidence

Top similarity score:

$$R = \frac{R_{raw} - r_{min}}{r_{max} - r_{min}} \quad (6)$$

Dataset statistics: $r_{min} = 0.3365$, $r_{max} = 0.6508$.

III. METHODOLOGY

A. Dataset and Distribution Strategy

1) *Source Data*: 50 labeled route-assistance queries stored in `test_queries.jsonl`. Binary labels indicate correctness.

2) *Experimental Setup*: No training; fixed 50-item evaluation. Retrieval corpus consists of synthetic road-notice documents.

B. Model Architecture

Pipeline:

Query $\rightarrow$ Retriever $\rightarrow$ LLM $\rightarrow$ Entropy Extraction $\rightarrow$ Fusion $\rightarrow$ Safety Gate.

C. Training Protocol

No parameter learning. Fusion weights fixed: $\alpha = 0.4$, $\beta = 0.6$, $\text{scale}=6.0$.

D. Experimental Variables

Compared:

- Polygraph-only uncertainty
- Fused uncertainty (retrieval + generation)

IV. RESULTS AND ANALYSIS

A. Performance Evaluation

TABLE I
UNCERTAINTY QUALITY METRICS

Method	AUROC	AUPRC	ECE
Polygraph-only	0.5246	0.6633	0.0404
Fused	0.4975	0.6690	0.2077

B. Trust Distribution

C. Risk-Coverage Analysis

At $\tau \geq 0.8$, coverage = 0.64, error = 0.375.

D. Calibration Analysis

Polygraph-only achieves lower ECE (0.0404) but yields compressed trust distribution unsuitable for gating.

Although the polygraph-only model shows good calibration, it generates compressed trust distributions that are not appropriate for decision gating.

Though the fused technique demonstrates overconfidence in mid-confidence bins, it also displays larger trust variability, indicating the necessity for post-hoc calibration.

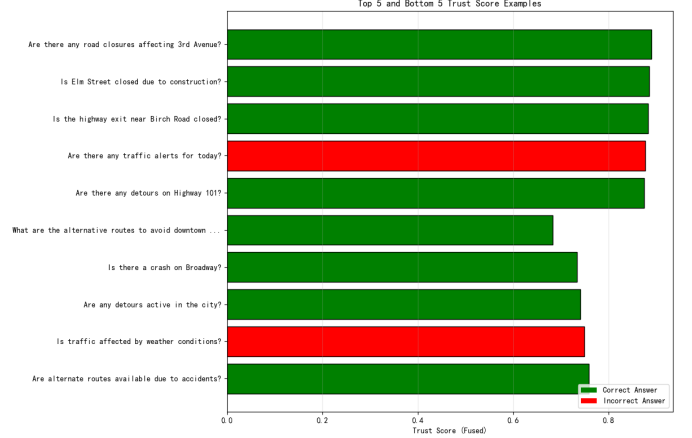


Fig. 1. Top and bottom trust score examples.

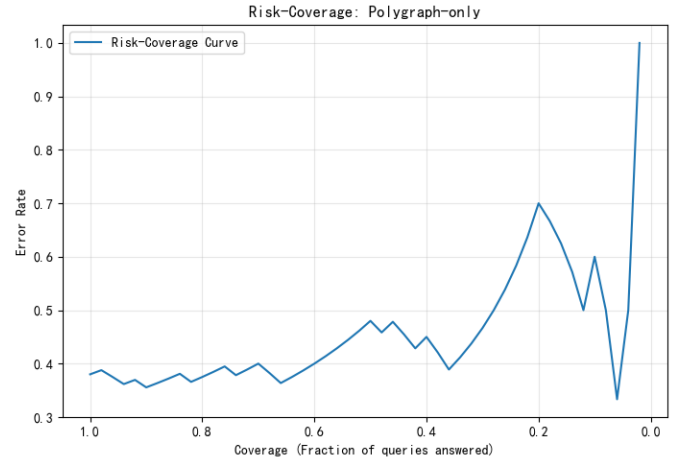


Fig. 2. Risk-coverage curves for polygraph-only models.

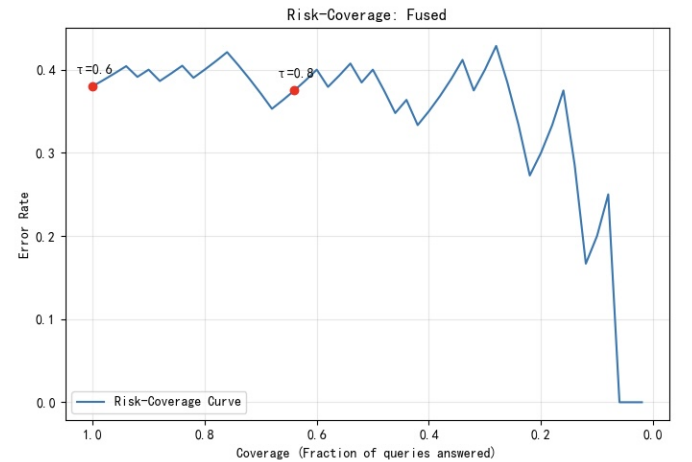


Fig. 3. Risk-coverage curves for fused models.

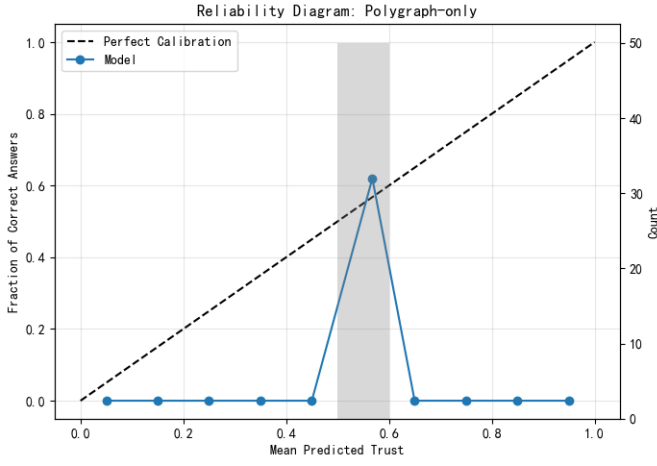


Fig. 4. Reliability diagram for polygraph-only uncertainty.

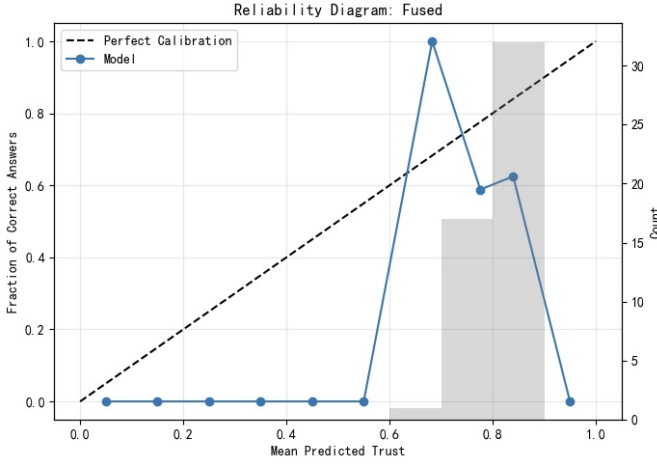


Fig. 5. Reliability diagram for fused uncertainty.

VI. REFERENCES

REFERENCES

- [1] O. Shorinwa et al., “A Survey on Uncertainty Quantification of Large Language Models,” arXiv:2412.05563, 2024.
- [2] C. Sharma, “Retrieval-Augmented Generation: A Comprehensive Survey,” arXiv:2506.00054, 2025.
- [3] D. Bouchard and M. S. Chauhan, “Uncertainty Quantification for Language Models,” arXiv:2504.19254, 2025.
- [4] E. Fadeeva et al., “Faithfulness-Aware UQ for RAG,” arXiv:2505.21072, 2025.
- [5] H. Soudani et al., “Uncertainty Quantification for Retrieval-Augmented Reasoning,” arXiv:2510.11483, 2025.

E. Key Observations

- Fusion improves operational coverage compared to polygraph-only.
- Calibration remains the primary challenge for heterogeneous signal fusion.

V. CONCLUSION

A. Summary of Findings

This work demonstrates that multi-source uncertainty fusion enables safety-aware gating in RAG-based route assistance systems. Although calibration requires improvement, fused trust scoring yields usable decision boundaries.

B. Future Extensions

- 1) Learned fusion via small neural calibration models.
- 2) Large-scale evaluation on real municipal datasets.
- 3) Adaptive thresholding using risk–coverage optimization.